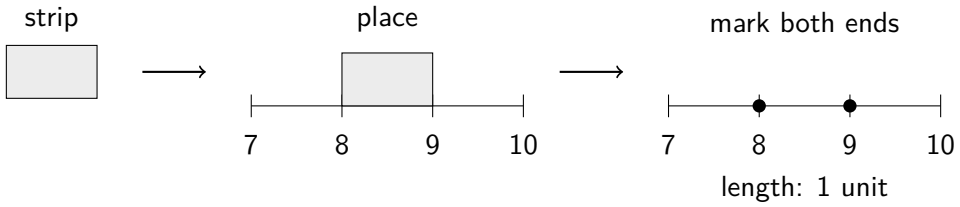
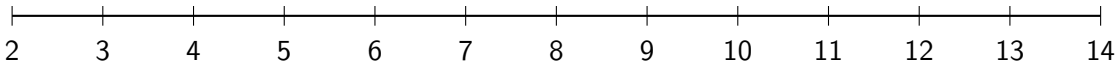
Use the four loose strips and the uncut row of squares. Each square is one unit. Lay strips along the ruler with both ends at printed marks.

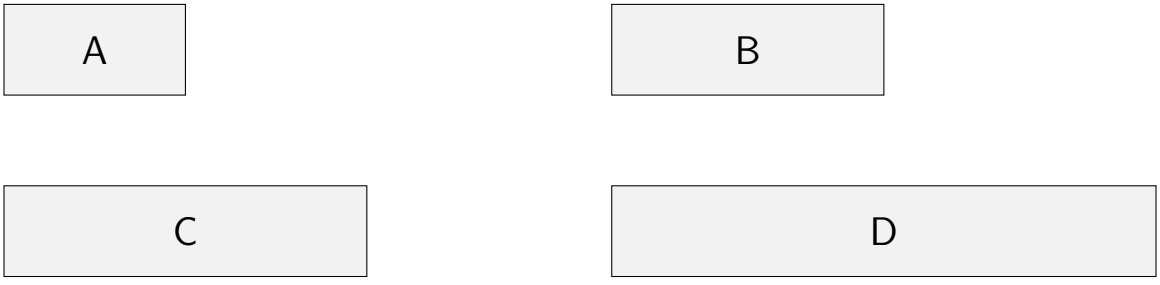


Problem 1: How many units long is each strip? Put each strip in different places on the ruler.



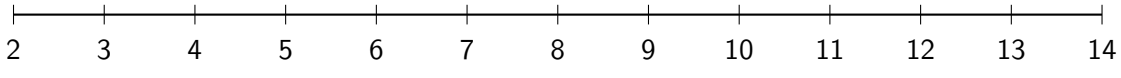
A: _____ B: _____ C: _____ D: _____



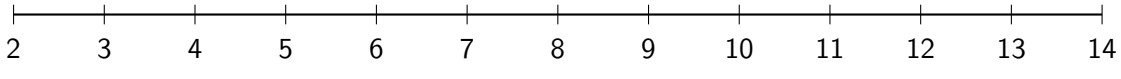


Problem 2: Find every place each strip can go with one end at 7. Draw its two ends on a ruler.

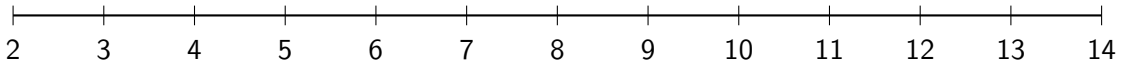
A



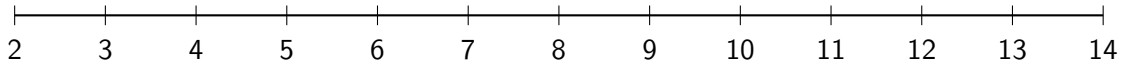
B



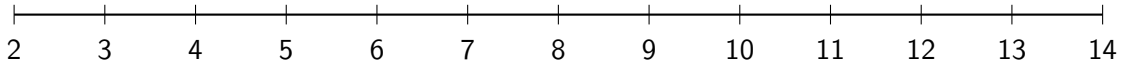
C



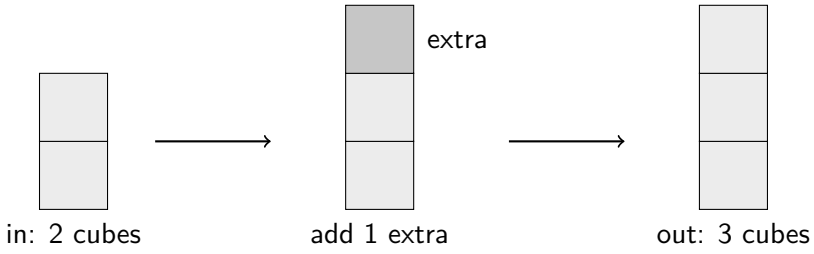
D



Problem 3: Which two strips can you put end to end and cover exactly with a third strip? Find every way.



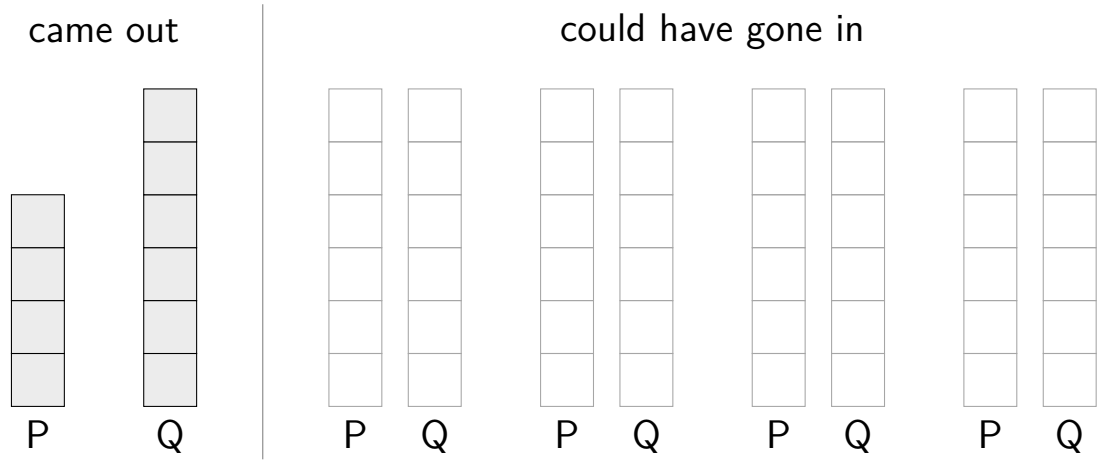
Behind a cover, the machine takes a tower of cubes and gives back a tower with the same number of cubes plus a secret extra. A machine always adds the same extra: 1, 2 or 3 cubes.



Problem 4: Take turns being the machine, with your secret extra as a stick of cubes behind the cover. Your partner finds the extra; then lift the cover to check.

Round	Towers you gave	Extra you found
1		
2		
3		
4		

Problem 5: P and Q went into the same machine: build and color every pair of towers that could have gone in. Could P be taller than Q?



Problem 6: P and Q still come out as 4 and 6 cubes, but each machine has its own extra. Build P and Q for each machine, and color them.

